

Qatar-9b

R-2518

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-9_250108 · created 2026-08-25T00:53:25+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460683.7506 +/- 0.0089 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.049
Transit depth (Rp/Rs)^2	1.99 +/- 1.39 [%]
Orbital Inclination (inc)	89.93 +/- 1.62
Airmass coefficient 1 (a1)	0.966 +/- 0.03
Airmass coefficient 2 (a2)	0.013 +/- 0.018
Scatter in the residuals of the lightcurve fit is	3.08 %
Best Comparison Star	#3 - [92, 375]
Optimal Method	PSF photometry
Transit Duration (day)	0.0714 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.141 ± 0.049 vs 0.1489 (deviation 0.16σ)
Duration fit vs published	0.0714 d vs 0.0779 d (deviation 0.99σ)
Mid-time O-C	-2.8 min
Depth SNR	2.9
Residual scatter	3.08 %

Light curve

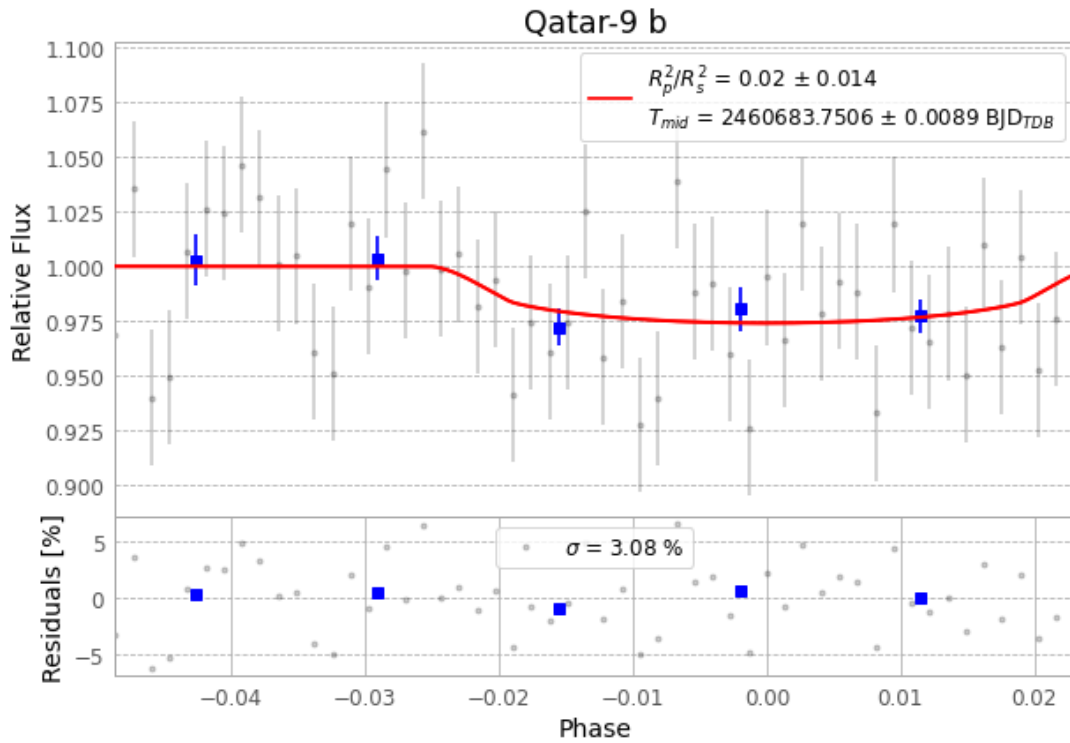


Figure 1 — FinalLightCurve_Qatar-9 b_2025-01-07.png (SHA-256 7945c4daa1e9a5c6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [407, 144], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c47a2349787cf2e0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

54 FITS frames (35 MB), Qatar-9250108040615.FITS → Qatar-9250108064515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-9-250108.log (727aab9b78f8...), FinalLightCurve_Qatar-9 b_2025-01-07.png (7945c4daa1e9...), FinalLightCurve_Qatar-9 b_2025-01-07.csv (852c5bb7ac6e...)

Compute resources

Wall time 97.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 00:53Z.